

GRID-CLOCK QUANTUM GRAVITY IV

A Finite Strict-Local Quantum Cellular Automaton for Emergent Spacetime, Einstein Gravity, Regular Black Holes, and Unitary Evaporation

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SCOPE NOTE. This revision reorganizes Paper IV around one candidate microscopic law. The core claim is not that quantum gravity has been completed, but that the previously separate BCC propagation, finite-Weyl geometry, Compute-Link capacity, emergent ADM/DeWitt sector, regular-core branch, and horizon information map can be embedded in one finite, unitary, strict-local QCA architecture. Exact statements, infrared derivations, conditional completions, and open tests are separated throughout.

Abstract

Grid-Clock Quantum Gravity (GCC) is developed here as a finite-information candidate architecture in which a fixed relational BCC graph and a universal microscopic update index are more fundamental than continuum spacetime. The principal advance of this revision is the construction of a single candidate many-body update, denoted U_{GCC}^{full} , acting on finite geometry, matter, frame-Link, load-Link, Compute, and reversible work registers. The update is formulated as an autonomous partitioned QCA so that unitarity and one-neighborhood-per-update causality are microscopic requirements rather than infrared approximations.

The uniform low-load sector reduces to the doubled-Weyl BCC propagator and therefore possesses an emergent relativistic light cone. Geometry records control the local QCA principal symbol; in the smooth sector the resulting dispersion relation is that of a Weyl/Dirac field in an ADM geometry. The same geometry-controlled unitary necessarily acts on the conjugate geometry record, giving a finite-dimensional backreaction identity whose low-phase limit is the Hamiltonian force generated by the matter Hamiltonian. A finite-Weyl geometry update with a quadratic DeWitt kinetic sector and Link-derived curvature then has the Einstein ADM Hamiltonian as its target infrared universality class. Requiring the standard normal-normal deformation bracket without an additional trace constraint selects the DeWitt coefficient one half, while the kinetic and curvature normalizations are reciprocal. The absolute Newton scale remains matched to the GCC Planck calibration rather than independently predicted.

At high load, finite cell capacity and the explicit reversible Compute-Link circuit bound the density by the Planck scale. In spherical symmetry the full correlated circuit yields a closed radial load equation; the uncorrelated limit reproduces the Hayward-type mass profile, while finite correlations generate bounded ultraviolet deformations that leave the leading exterior correction unchanged. Frame holonomy and scalar load are represented by distinct Link factors, preventing an incorrect identification of curvature phase with load phase.

A horizon is not a fundamental gate or defect. When a semiclassical state is cut into inside and outside sectors, the orientation-independent normal-current projector of the same four-component bulk carrier has rank two, producing an effective M_2 boundary algebra. Conditional on boundary-support minimality, gauge-reduced counting gives one binary boundary record per independent patch. The resulting horizon tower admits an explicit finite-dimensional unitary information-transfer map. Microcanonical reciprocity fixes inclusive emission relative to the QCA absorption probability, so no independent horizon-source normalization is introduced. The full architecture is therefore a sharpened quantum-gravity candidate, not a completed theorem: a concrete finite update has been structurally specified, but full interacting-field closure, controlled renormalization, dynamical collapse/rotation, greybody calculations, and a complete many-body physical-state construction remain decisive tests.

Keywords: quantum gravity; quantum cellular automaton; finite information; finite Weyl algebra; BCC lattice; emergent Lorentz symmetry; ADM/DeWitt gravity; regular black holes; unitary evaporation; Page curve.

Claim-status summary

Element	Status	Meaning
Fixed BCC graph and universal update index	Foundational postulates	Microscopic adjacency and update ordering are fundamental; continuum spacetime is emergent.
Finite local Hilbert spaces	Explicit quantum postulate	Geometry, matter-control, Link, and work records are finite-dimensional.
Finite local capacity	Foundational completion postulate	Bounds localized energy and fixes the ultraviolet density scale.
U_GCC^full partitioned QCA	Candidate microscopic completion	A single autonomous strict-local unitary architecture is specified structurally; a unique finite matrix realization is not yet selected.
BCC/Weyl free propagation	Explicit candidate sector	Exact radius-one unitary; relativistic cone emerges at low wave number.
Curved-space matter sector	Conditional infrared derivation	Geometry-controlled local unitaries yield the ADM light cone and Weyl/Dirac principal operator.
Matter backreaction	Exact finite identity + IR limit	Geometry-controlled unitary acts reciprocally on the conjugate geometry record.
ADM/DeWitt gravity	Infrared universality target	DeWitt coefficient is selected conditionally by closure; H-H/H-D/D-D have analytic/numerical support in the smooth sector.
Newton constant	Matched, not derived	Absolute normalization is tied to the Planck calibration; no dimensionless prediction is claimed.
Link holonomy curvature	Candidate microscopic realization	Frame-Link loops reconstruct curvature in the smooth sector; strict locality requires multi-tick accumulation.
Planck-density core	Robust capacity result	Density bound and cube-root core scaling precede the detailed constitutive law.
Hayward-type profile	Conditional branch	Exact for the minimal uncorrelated quadratic closure; correlated full circuit gives a bounded deformation.
Rank-two boundary algebra	Exact kinematic result	Orientation-independent outgoing support algebra is isomorphic to $M_2(\mathbb{C})$.
Binary horizon record	Conditional gauge reduction	Requires boundary-support minimality; no raw BCC edge is identified with a qubit.
Unitary evaporation and detailed balance	Explicit candidate sector	Finite-dimensional transfer map and microcanonical normalization are explicit; greybody/exclusive amplitudes remain to be calculated.
Full interacting 3+1 theory	Open verification problem	Physical-state completeness, renormalization, collapse, rotation, and all-sector closure remain unproved.

1. Introduction

The central problem addressed by GCC is not simply how to quantize a classical metric. It is whether quantum matter, relativistic causal propagation, Einstein gravity, finite strong-gravity behavior, and black-hole information transfer can be understood as different sectors of one finite local update law. This revision treats that question as the organizing principle of Paper IV.

The microscopic substrate is a fixed graph. The graph does not stretch, contract, or acquire continuum curvature as a fundamental object. What changes are finite records attached to nodes and Links: local frame/geometry information, matter amplitudes, capacity/load variables, and reversible scheduling data. Continuum distance, proper time, connection, metric, and curvature are relational observables reconstructed from these records. In this sense GCC differs in microscopic interpretation from a continuum theory in which the spacetime metric itself is fundamental, while still requiring general relativity as its smooth low-energy universality class.

The revision is built around a strict distinction between three logical levels. First, the microscopic QCA must be finite, unitary, autonomous, and causal. Second, a controlled smooth/low-phase sector must reproduce relativistic matter and the complete ADM/DeWitt deformation algebra. Third, the finite-capacity ultraviolet sector must remain consistent with the same microscopic update rather than being introduced by a separate black-hole rule. Statements that have not reached the corresponding level are kept explicitly conditional.

$$U_{GCC}^{full} \rightarrow \left(\begin{array}{l} \text{Weyl/QFT and Lorentz symmetry} \\ \text{Einstein gravity in the smooth IR} \\ \text{regular strong gravity at saturation} \\ \text{unitary black-hole information transfer} \end{array} \right)$$

The rest of the paper constructs each arrow in this sector map, identifies where exact finite identities are available, and states the remaining verification program.

PART I. ONE FINITE MICROSCOPIC LAW

2. Fixed graph, universal update, finite information, and capacity

GCC assumes a fixed relational BCC graph and a universal microscopic update label K . Coordinate labels and graph embeddings are auxiliary; only adjacency and relational records enter the microscopic law. The update label is an ordering variable, not directly an observable proper time. Local proper time and continuum lapse are emergent record-dependent quantities.

Finite information is implemented by finite-dimensional local Hilbert spaces. Finite energy capacity is a distinct postulate: a four-component propagation carrier does not imply four independent Planck-energy slots. This separation corrects the earlier Minimal Action Saturation interpretation and prevents Hilbert-space dimension from being misread as local energy capacity.

$$Z X = \omega X Z, \omega = e^{2\pi i/d}$$

Finite Weyl clock/shift pairs provide exact unitary record translations. Bounded Hermitian functions of these finite records are used for geometry, load, and local physical observables. The exact microscopic algebra is therefore finite; canonical commutators arise only in low-phase charts.

$$E_{cell} \leq E_p, \rho \leq \rho_p$$

When one BCC processing node is identified with one GCC capacity cell at Planck energy and Planck density, the relational processing scale satisfies the following calibration.

$$4 \ell_{proc}^3 = \ell_p^3, \ell_{proc} = 4^{-1/3} \ell_p, t_* = 4^{-1/3} t_p$$

This calibration fixes a branch of the microscopic scale dictionary. It does not turn the four propagation amplitudes into four independent energy capacities.

3. Full local Hilbert factorization

The minimal integrated architecture separates geometry, matter, capacity support, and reversible scheduling at nodes, and separates frame transport from scalar load on edges. This avoids the inconsistency that would result from using one Link phase simultaneously as a spin connection and as a strong-gravity load variable.

$$H = \bigotimes_x (H_{G_x} \otimes H_{M_x} \otimes H_{C_x} \otimes H_{W_x}) \otimes \bigotimes_e (H_{L_e^{frame}} \otimes H_{L_e^{load}})$$

The geometry register G contains a finite inverse-triad-like record and its finite Weyl conjugate refresh. The matter register M contains the strict-local propagation carrier and any additional internal species records. The Compute register C carries local support/capacity information. The work register W stores reversible scheduler, memory, and temporary holonomy accumulation data. Each edge has a frame Link, which compares local frames and transports matter, and a load Link, which carries the finite-Weyl scalar mismatch used by the Compute-Link saturation sector.

4. Autonomous strict-local QCA

The fundamental law is a discrete-time unitary, not the exponential of a local Hamiltonian. A finite-time exponential of a nearest-neighbor Hamiltonian generically develops longer-range tails, whereas the microscopic GCC postulate requires a hard one-neighborhood causal cone per update.

$$|\Psi_{K+1}\rangle = U_{GCC}^{full} |\Psi_K\rangle$$

To combine several local tasks without allowing information to propagate through multiple freshly updated neighborhoods inside one K-step, the QCA is partitioned and double-buffered. An internal scheduler register makes the update autonomous.

$$U_{GCC}^{full} = \sum_{s=0}^{R-1} |s+1\rangle \langle s|_W \otimes \left(\prod_{B \in P_s} u_B \right)$$

Within each scheduler phase the blocks in the partition are disjoint. Every local gate reads records from the current slice and writes the next slice; freshly written records are not reused as controls until the next update. Consequently a local Heisenberg observable can spread by at most one graph neighborhood per microscopic step.

$$\text{supp}(U^\dagger O_x U) \subseteq A_{N_1(x)}$$

The scheduler is not an external clock program. Its phase is a finite internal record shifted by the same autonomous law. Several microscopic ticks may form a convenient Floquet supercycle for infrared analysis, but the fundamental law remains one repeated QCA update.

5. Two levels of exactness and microscopic gauge

A finite commutative site algebra admits no nontrivial ordinary Leibniz derivation. GCC therefore does not demand the continuum infinitesimal hypersurface-deformation algebra as an exact operator identity at a fixed finite cutoff. Instead it assigns exactness to finite matrix identities, gauge-covariant update composition, and finite groupoid/path consistency. The ADM/DeWitt algebra is an emergent smooth-sector requirement.

Tangential deformation on the fixed graph is interpreted as relational address relabeling rather than literal graph motion. The finite triad record also has an internal frame redundancy because the metric is invariant under a local rotation of the triad. These redundancies define the microscopic gauge sector. A candidate physical subspace is obtained by projecting or averaging over the frame and address gauge actions.

$$P_{frame} = \int dg U_{frame}(g), P_{addr} = \frac{1}{|G_{addr}|} \sum_{\gamma \in G_{addr}} U_{addr}(\gamma)$$

$$P_{phys} = P_{addr} P_{frame}, H_{phys}^{micro} = P_{phys} H, [U_{GCC}^{full}, P_{phys}] = 0$$

If the full update is gauge covariant, the physical projector commutes with the QCA and the restricted evolution remains unitary. A gauge-singlet uniform vacuum shows that the projected space need not be empty. This is a construction route, not a classification theorem for the complete 3+1 physical Hilbert space.

PART II. MATTER AND EMERGENT RELATIVITY

6. Uniform vacuum: the strict-local BCC/Weyl sector

In a uniform, low-load geometry the full update must reduce to a homogeneous propagation QCA. The preferred minimal carrier is the four-component doubled-Clifford BCC walk. Its eight neighbor directions are the sign triples of the three spatial axes.

$$\alpha_i = \sigma_x \otimes \sigma_i, P_i^\pm = \frac{I \pm \alpha_i}{2}, A_s = P_x^{s_x} P_y^{s_y} P_z^{s_z}$$

$$\psi_{K+1}(n) = \sum_{s \in \{\pm 1\}^3} A_s \psi_K(n-s)$$

The update is a single radius-one convolution. It is not interpreted as three sequential physical shifts. In Fourier space the same candidate can be written as a product of finite Clifford rotations.

$$U(q) = e^{-iq_x \alpha_x} e^{-iq_y \alpha_y} e^{-iq_z \alpha_z}$$

After a fixed carrier-basis rotation, the four-component walk decomposes into two Weyl sectors of opposite chirality. Published classifications of homogeneous, local, isotropic, two-component walks in three dimensions select the BCC Cayley geometry and the two Weyl walks within the stated class. GCC uses the doubled carrier because it provides the minimal four-component structure employed here for two positive-frequency propagation amplitudes. The classification does not exclude more general nonlinear or higher-dimensional QCA completions.

7. Geometry-controlled propagation and the curved-space causal cone

The geometry record contains a finite triad-like variable. In a smooth semiclassical state it defines the spatial metric and inverse triad.

$$h_{ab} = E_a^i E_b^i, h^{ab} = e_i^a e_i^b$$

The local lapse and shift are read from the same geometry/scheduler record family. The matter QCA is not curved by moving the graph. Instead, the local scattering/coin and Link transport are conditioned on the geometry record. Its principal velocity matrix is chosen to approach the following smooth expression.

$$V^a[G] = N^a I + c N e_i^a \alpha^i$$

Using the Clifford anticommutation relations, the local dispersion satisfies the ADM null-cone relation.

$$(\omega - N^a k_a)^2 = c^2 N^2 h^{ab} k_a k_b$$

Thus a fixed microscopic graph can support an emergent curved-spacetime causal cone. The frame Links supply the covariant transport connection, so the low-energy one-particle generator has the Weyl/Dirac principal form.

$$H_M^{IR}[G, L] = \frac{1}{2} (N^a I + c N e_i^a \alpha^i, -i \hbar D_a) + H_{mass} + H_{int}$$

This establishes the kinematic curved-QFT branch conditionally: the exact finite QCA is fundamental, while the displayed differential operator is its smooth low-phase generator. Interacting many-body QFT requires additional local species and interaction gates and remains part of the verification program.

8. Unitary matter backreaction from the same coupling

A separate classical source rule is unnecessary at the architectural level. Let a finite geometry basis control a matter unitary.

$$C_{GM} = \sum_g |g\rangle \langle g| \otimes U_M(g)$$

Because the conjugate geometry update is a finite shift, conjugating that shift by the same controlled matter unitary gives an exact finite-dimensional identity.

$$C_{GM}^\dagger (X_G \otimes I) C_{GM} = X_G \Omega_G, \Omega_G = \sum_g |g\rangle \langle g| \otimes U_M(g+1)^\dagger U_M(g)$$

In a low-phase chart, where the finite shift is represented by the exponential of a conjugate momentum and the matter update by a short effective Hamiltonian step, the identity reduces to the ordinary Hamiltonian backreaction force.

$$\Delta p_A = -t \cdot \frac{\partial H_M}{\partial q^A} + O(t^2)$$

The key point is reciprocity: geometry controls matter propagation, and the same unitary necessarily updates the conjugate geometry record according to the geometry dependence of the matter dynamics. The matter stress tensor is therefore not an unrelated second postulate. In the continuum metric sector this is consistent with the standard metric variation of the matter action. A complete proof of universal coupling across all matter species is still required.

PART III. EMERGENT EINSTEIN GRAVITY

9. Finite geometry dynamics

The geometry update has a bounded kinetic part, a curvature part derived from frame Links, and an address/shift part. In the smooth chart the effective Hamiltonian is required to approach the most general local quadratic DeWitt kinetic form with a scalar-curvature potential.

$$H_G[N, N^a] = \int d^3x \left\{ N \left[\frac{A}{\sqrt{h}} (\pi^{ab} \pi_{ab} - \lambda \pi^2) - B \sqrt{h} R^{(3)} \right] + N^a D_a \right\}$$

At the microscopic level the momentum variables are bounded finite-Weyl functions rather than unbounded canonical operators. The displayed Hamiltonian is therefore an infrared generator of a finite unitary supercycle, not a fundamental continuous-time Hamiltonian.

10. Closure selection of the DeWitt coefficient

The DeWitt trace coefficient is not left arbitrary. In the continuum bracket of two normal generators, the cross-term between the general quadratic kinetic term and the scalar-curvature potential produces the momentum constraint plus an additional trace-gradient term.

$$\{H_\lambda[N], H_\lambda[M]\} = D[\omega] + (1 - 2\lambda) \int d^3x (\nabla_a \pi) (N \nabla^a M - M \nabla^a N)$$

For generic states, if no additional trace constraint is introduced, the standard hypersurface-deformation algebra requires the coefficient of the extra term to vanish. Matching the normalization of the remaining momentum constraint also fixes the reciprocal kinetic/curvature product.

$$\lambda = \frac{1}{2}, AB = 1$$

The resulting target infrared Hamiltonian density is the ADM gravitational Hamiltonian with an effective Newton constant.

$$H_G = \frac{16\pi G_{eff}}{\sqrt{h}} \left(\pi^{ab} \pi_{ab} - \frac{1}{2} \pi^2 \right) - \frac{\sqrt{h}}{16\pi G_{eff}} R^{(3)}$$

The selection is conditional on the assumed local quadratic kinetic sector, the scalar-curvature potential, and the absence of an additional trace constraint. Within that sector it sharpens the earlier numerical observation that the closure root converges to the general-relativistic value.

11. Frame-Link holonomy and emergent spatial curvature

The fixed graph does not acquire microscopic curvature. Curvature is reconstructed from non-integrable frame transport. For each oriented BCC edge, the frame Link transforms covariantly under local frame rotations.

$$L_{x,s} \rightarrow G_x L_{x,s} G_{x+s}^\dagger, L_{x+s,-s} = L_{x,s}^\dagger$$

A minimal BCC four-edge loop provides a finite holonomy.

$$W_{x;s,t} = L_{x,s} L_{x+s,t} L_{x+t,s}^\dagger L_{x,t}^\dagger$$

In a smooth low-phase chart, the loop holonomy measures the curvature two-form through its area bivector.

$$W_{x;s,t} = \exp \left[-\frac{i}{4} \Sigma_{s,t}^{ab} F_{ab}^i \sigma_i + O(\ell_{proc}^3) \right]$$

Independent BCC loop bivectors span the three independent spatial two-form directions. Contracting the recovered frame curvature with the inverse triad gives the scalar curvature required by the ADM potential.

$$R^{(3)} = \epsilon_k^{ij} e_i^a e_j^b F_{ab}^k$$

A subtle locality point is essential. A node cannot instantaneously read all four Links of a loop whose opposite corner lies two graph steps away. GCC therefore uses a reversible work register to accumulate the loop product

over several microscopic ticks. Each tick remains radius one. The resulting curvature cache is retarded by only a Planck-scale number of ticks and agrees with the instantaneous curvature in the smooth limit up to derivative-suppressed corrections.

12. Infrared hypersurface-deformation algebra

The geometric origin of the H-H structure function can be seen directly from the response of a unit normal to a normal deformation. Orthogonality to the tangents gives an inverse-metric-weighted lapse gradient.

$$\delta_N n = -h^{ab}(\partial_b N) e_a$$

The order difference of two normal deformations is therefore tangential and has the standard inverse-metric structure.

$$[\delta_N, \delta_M] X = e_a h^{ab} (N \partial_b M - M \partial_b N)$$

At the canonical level this is the target normal-normal bracket.

$$\{H[N], H[M]\} = D[h^{ab} (N \partial_b M - M \partial_b N)]$$

The existing numerical audits support the distinction between full-cutoff and smooth-sector closure. Full-cutoff H-D and D-D residuals remain order unity under refinement, whereas fixed low Fourier modes converge with approximately second-order accuracy. The normal-normal finite-path coefficient also approaches unity as the deformation step is reduced.

epsilon	Raw H-H diamond	Residual after fitted shift	Fitted coefficient
0.040	3.17×10^{-4}	2.52×10^{-6}	0.99552
0.020	7.93×10^{-5}	3.01×10^{-7}	0.99723
0.010	1.98×10^{-5}	3.49×10^{-8}	0.99810
0.005	4.96×10^{-6}	4.16×10^{-9}	0.99853

Sites	a	Low-mode H-D residual	Low-mode D-D residual
32	0.19635	0.11249	0.17904
64	0.09817	0.02875	0.04623
128	0.04909	0.007227	0.01165
256	0.02454	0.001809	0.002918

These results are evidence for the required infrared universality class, not a proof that every nonlinear many-body sector of the proposed full QCA is anomaly-free.

13. Einstein-Hilbert reconstruction and normalization status

If the geometry, matter, and gauge sectors described above share the same smooth lapse and shift, if the full H-H/H-D/D-D algebra closes in the refined physical sector, and if no additional low-energy gravitational degree of freedom survives, the effective action takes the Einstein-Hilbert plus matter form with higher-derivative ultraviolet corrections.

$$S_{eff} = \frac{c^3}{16\pi G_{eff}} \int d^4x \sqrt{-g} R + S_{matter} + \Delta S_{UV}$$

The relative normalization between the DeWitt kinetic term and curvature term is constrained by closure. The absolute value of the effective Newton constant is not independently derived in the present construction. It is matched to the GCC Planck calibration. A future microscopic renormalization calculation would have to show that this matching is stable and unique rather than a phenomenological identification.

PART IV. FINITE CAPACITY AND REGULAR STRONG GRAVITY

14. Capacity saturation and robust core scaling

No elementary capacity cell is required to represent arbitrarily large localized energy. The density bound alone implies a minimum volume for a fixed enclosed mass. In spherical symmetry this gives the robust cube-root core scale.

$$r_c = \left(\frac{3M}{4\pi\rho_p} \right)^{1/3} \propto M^{1/3}$$

This scaling does not require a Hayward metric, a particular Link harmonic, or an equal Compute-Link load assumption. It follows from finite density, three-dimensional volume counting, and finite total mass. The detailed radial profile requires an additional constitutive closure.

15. Explicit Compute-Link circuit and the correlated radial equation

The earlier quadratic strong-gravity branch can be realized by two required local support roles: Compute and Link. The revised architecture retains the explicit reversible correlation circuit rather than replacing its quantum correlations by an arbitrary radial interpolation. Using the mean enclosed density as the coarse load gives the dimensionless variable below.

$$\chi(r) = \frac{3m(r)}{4\pi\rho_p r^3}$$

After tracing only reversible work registers, the explicit Compute-Link circuit gives the normalized density response

$$f_g(\chi) = \frac{\rho}{\rho_p} = \chi^2 + \chi(1-\chi)\sin^2(g\chi^2)$$

The response is bounded and has exact vacuum and saturation endpoints.

$$0 \leq f_g(\chi) \leq 1, f_g(0) = 0, f_g(1) = 1$$

Combining mass conservation with the definition of the mean-density load yields a closed radial equation for the full correlated circuit.

$$r \frac{d\chi}{dr} = -3\chi(1-\chi)\cos^2(g\chi^2)$$

This equation improves the logical status of the strong-gravity branch. The radial dependence is no longer freely chosen once the circuit response is specified. The remaining completion choice is the microscopic identification between the scalar Link load and the coarse spherical load.

16. Hayward limit, correlated deformations, and exterior behavior

In the uncorrelated limit the radial equation integrates exactly.

$$g=0: \chi(r) = \frac{a^3}{r^3+a^3}, m(r) = \frac{Mr^3}{r^3+a^3}, a^3 = \frac{3M}{4\pi\rho_p}$$

This is the Hayward-type mass profile used in the earlier revision. For nonzero circuit correlation the exact Hayward form is deformed, but the density cap and central saturation remain intact. Near the vacuum the correlation correction enters only at fifth order in the load; therefore the leading $p=2$ exterior correction remains quartic in inverse radius, while the new correlation-dependent correction appears at much higher inverse power. The detailed coefficient is microscopic-model dependent.

At the center the load approaches unity and the density approaches the Planck value. The corresponding static spherical geometry is de Sitter-like rather than singular. This establishes a regular strong-gravity branch conditionally on the chosen local constitutive dictionary; it does not prove dynamical collapse into that branch or stability of an inner horizon.

17. Frame curvature and scalar load must be distinct Link factors

A key consistency correction is that the frame-holonomy phase of Section 11 cannot be identified directly with the scalar finite-Weyl load phase. In the weak-field Einstein sector, curvature is proportional to energy density; in the minimal two-channel branch the density is quadratic in the load. Therefore the curvature holonomy begins quadratically in the scalar load, not linearly.

$$H_L = H_{L^{frame}} \otimes H_{L^{load}}$$

$$\theta_{curvature} \propto f_g(\chi) = \chi^2 + O(\chi^5)$$

The two Link factors remain parts of one neighboring-record architecture. The frame factor transports local orientation and reconstructs curvature; the scalar factor carries the bounded capacity mismatch. A controlled source gate can use the bounded Compute-Link support to rotate the frame sector, connecting load to curvature without equating the two phases.

PART V. HORIZONS AND UNITARY INFORMATION TRANSFER

18. The horizon is an emergent cut, not a microscopic defect

The full QCA contains no horizon flag. Event-horizon membership is a global causal property, while the local curvature at the horizon of a very massive black hole can be small. A special surface coupling would therefore introduce a nonlocal retuning signal. In the minimal GCC branch, the same local update acts everywhere; a horizon is identified only after a semiclassical state is partitioned into inside and outside causal sectors.

19. Orientation-independent rank-two boundary algebra

Raw BCC edge counting is regulator dependent and fails as a universal horizon count. The correct kinematic object is the normal-current split of the same four-component bulk carrier. For any physical unit normal, the Clifford algebra gives two outgoing and two ingoing eigenchannels.

$$P_{out} = \frac{I + \alpha_{\hat{n}}}{2}, P_{in} = \frac{I - \alpha_{\hat{n}}}{2}, \text{rank } P_{out} = \text{rank } P_{in} = 2$$

The outgoing support algebra is therefore

$$A_H = P_{out} M_4(C) P_{out} \simeq M_2(C)$$

This result is orientation and frame independent because local frame rotations act by unitary conjugation. It is an exact kinematic statement about the bulk carrier. It is not, by itself, a theorem that the full many-body horizon Hilbert space contains no additional independent memory.

20. Boundary support minimality, counting, and area matching

The remaining counting assumption can be sharpened as boundary-support minimality: after the inside/outside split and gauge reduction, independent horizon memory is exhausted by the outgoing support algebra of the universal bulk carrier; species and polarization labels are scattering channels rather than independent surface-memory factors. Under this condition the minimal irreducible boundary record is binary.

For a closed connected horizon, one global binary consistency condition reduces the raw count by one factor of two.

$$\Omega_F = 2^{F-1}, S_H = k_B (F - 1) \ln 2$$

Finite-Weyl kinematics fixes the entropy increment of one independent record but not its physical area. The conversion is fixed by matching to the required semiclassical stationary-horizon first law and Hawking temperature. One binary record then corresponds to

$$\Delta A = 4 \ln 2 \ell_p^2$$

The area coefficient is therefore a micro-IR matching result, not a pure finite-state counting theorem.

21. Discrete horizon tower and exact finite information transfer

Using the regular extremal geometry as an information-free offset, the horizon information sectors may be organized by the number of independent binary records.

$$A_n = A_{ext} + 4n \ln 2 \ell_p^2, S_n = k_B n \ln 2$$

The crucial dimension identity for a transition that transfers q records to radiation is exact.

$$\dim H_H(n) = 2^n = 2^{n-q} 2^q = \dim [H_H(n-q) \otimes H_R(q)]$$

Therefore each energy-compatible sector admits a unitary isomorphism between the initial horizon space and the smaller horizon space tensored with the emitted record space. For one record, the explicit map used in Paper IV is

$$|g_n; b_1, \dots, b_n\rangle |0\rangle_R \mapsto |g_{n-1}; b_1 \oplus b_n, b_2, \dots, b_{n-1}\rangle |b_n, \epsilon_n\rangle_R$$

$$M_n c^2 = M_{n-1} c^2 + \epsilon_n$$

The map is bijective and extends to an exact finite-dimensional unitary. Repeated application transfers progenitor boundary information to radiation/history correlations without requiring the final geometric endpoint to carry a large information remnant. In the full theory this map is not a special horizon gate; it is the effective block of the same universal QCA restricted to the corresponding black-hole/radiation sector.

22. Microcanonical reciprocity and inclusive Hawking normalization

Because the microscopic law is reversible, the inclusive emission and absorption matrix-element sums differ only by the number of initial black-hole microstates. For a transition removing q independent records,

$$\frac{P_{emit}}{P_{abs}} = \frac{\Omega_{n-q}}{\Omega_n} = 2^{-q}$$

If the no-boundary-defect branch is used, the absorption probability is the greybody probability of the same exterior QCA. The inclusive emission probability is therefore

$$P_{emit}(n \rightarrow n-q; \alpha) = 2^{-q} T_\alpha(\omega_{nq})$$

No independent phenomenological horizon-source constant is required. The remaining unknowns are the exterior QCA scattering matrix, mode density and group velocity, and exclusive many-body amplitudes. In a smooth normalization the inclusive occupation takes the usual Bose/Fermi entropy form, with the QCA flux measure replacing the continuum one when necessary.

$$\frac{d^2 N}{dt d\omega} = \sum_\alpha \Phi_{QCA, \alpha}(\omega) \frac{T_\alpha(\omega)}{\exp[\Delta S_{BH}(E, \omega)/k_B] \mp 1}$$

23. Sector-centroid gaps, Page turnover, and quasienergy

The area tower gives a sector-centroid energy gap that approaches the Hawking temperature times one binary entropy unit for a large Schwarzschild black hole.

$$\epsilon_n \simeq k_B T_H \ln 2, \frac{\epsilon_n}{k_B T_H \ln 2} = 2n - 2\sqrt{n(n-1)}$$

The tower does not require exact degeneracy of all microstates at fixed n . Interactions in the full QCA may split the exponentially many microstates into a dense band, so a resolvable line spectrum is conditional rather than robust. The more robust prediction is the finite-sector entropy correction to detailed balance.

For a typical initially scrambled state, unitary record transfer gives the standard Page turnover.

$$S_R(n) \simeq k_B \ln 2 \min(n_0 - n, n), M_{Page} = \frac{M_0}{\sqrt{2}}, \frac{t_{Page}}{\tau} = 1 - \frac{1}{2\sqrt{2}}$$

Because the microscopic dynamics is discrete-time, exact conservation is most naturally formulated in terms of QCA quasienergy. For an R-tick autonomous supercycle,

$$U=\left(U_{GCC}^{full}\right)^R, H_{eff}=\frac{i\hbar}{Rt_*}\log U$$
$$\varepsilon\sim\varepsilon+\frac{2\pi\hbar}{Rt_*}$$

In the low-phase branch this quasienergy is identified with ordinary physical energy. The blockwise mass-gap relation should therefore be read as an exact sector matching within the chosen low-phase energy branch, not as a claim that an unbounded Hamiltonian is fundamental.

PART VI. INTEGRATED AUDIT AND VERIFICATION PROGRAM

24. One-law sector map

The construction is now organized around one microscopic architecture rather than a collection of independent models. The following table summarizes the sector conditions and the corresponding effective theory.

Sector / limit	Dominant records and update	Effective result	Current status
Uniform vacuum; low wave number	Matter carrier on uniform frame Links	Doubled Weyl QCA; emergent Lorentz cone	Explicit candidate
Smooth geometry; low load	Geometry-conditioned coin + frame transport	Curved Weyl/Dirac principal dynamics	Conditional IR derivation
Dynamical smooth geometry	Same matter coupling + finite geometry refresh	Matter backreaction and ADM/DeWitt target	Conditional; partial analytic/numerical closure
Refined physical sector	Gauge reduction + low-mode HDA	Einstein-Hilbert + matter universality class	Target; not a full theorem
High capacity load	Compute + load-Link finite circuit	Planck density bound; regular core	Robust bound + conditional profile
Semiclassical inside/outside cut	Normal-current projection of bulk carrier	Rank-two boundary algebra	Exact kinematic result
Evaporating BH sector	Same QCA restricted to horizon/radiation blocks	Unitary information transfer + microcanonical emission	Explicit candidate; scattering incomplete

25. What is exact, derived, conditional, and open

Exact finite statements include the finite Weyl algebra, the unitarity of the chosen local gates, the partitioned causal construction, the geometry-controlled backreaction identity, the rank of the normal-current projector, the M2 outgoing algebra, the horizon-sector dimension identity, and the bijectivity of the explicit information-transfer map.

Derived infrared statements include the ADM causal cone from the geometry-controlled principal symbol, the standard H-H structure from normal response, and the DeWitt one-half selection within the stated local quadratic sector. Existing numerical audits support H-D and D-D convergence only in a smooth low-mode refinement sector.

Conditional completions include the exact microscopic dictionary from finite geometry records to local tetrads and frame Links, the controlled coarse graining to the full Einstein-Hilbert action, the absolute Newton normalization, the scalar Link load dictionary, the positive-harmonic selection, the detailed regular mass profile, boundary-support minimality, and the embedding of all exclusive evaporation amplitudes into one interacting QCA.

Open problems include a unique explicit finite-dimensional realization of every local gate in the nonlinear 3+1 theory, controlled renormalization, a complete physical-state/groupoid-averaging construction, universal interacting matter coupling, dynamical collapse, rotating Kerr-like solutions, inner-horizon stability, the QCA greybody spectrum, microstate splittings/linewidths, and an interacting-field Page curve beyond the typical-state record model.

26. Falsifiability and consistency conditions

- No candidate microscopic update is acceptable if one K-step propagates information beyond the declared graph neighborhood.

- The refined low-energy sector must recover the complete H-H, H-D, and D-D algebra; success in only one bracket is insufficient.
- The same geometry record that controls matter propagation must generate compatible backreaction and must not require species-dependent retuning that violates universal coupling.
- The physical gauge-reduced Hilbert space must contain nontrivial excitations and be invariant under the full update.
- The refinement/renormalization map must suppress lattice Lorentz and diffeomorphism violations at observable scales.
- The finite-capacity branch must preserve the density bound and remain dynamically stable under collapse and rotation.
- The scalar Link load dictionary and harmonic assumptions must be derived, independently motivated, or empirically constrained.
- The boundary rank-two sector and information-transfer map must survive the full many-body construction without an additional nonlocal horizon flag.
- Once the exterior absorption problem is calculated, emission must obey the microcanonical reciprocity normalization rather than require a new fitted horizon coupling.

27. Decisive numerical verification program

The next stage should be computational rather than conceptual. A small periodic BCC lattice with finite node and edge registers can test the candidate architecture before attempting large-scale phenomenology. The first two diagnostics are exact matrix tests of unitarity and causal support.

$$\varepsilon_U = \| U^\dagger U - I \|, \varepsilon_{caus}(r>1) = \| \Pi_{r>1}(U^\dagger O_x U) \|$$

The second layer is a low-mode refinement audit. The free sector should reproduce the Weyl cone, the geometry-controlled sector should reproduce the ADM principal symbol, and the full constraint generators should converge to the HDA on a fixed physical Fourier band.

$$\varepsilon_{HDA}(a, k_{max}) \rightarrow 0, \varepsilon_{Lorentz}(k \ell_{proc}) \rightarrow 0$$

The third layer is nonlinear: prepare a localized matter packet and verify reciprocal geometry momentum transfer; evolve a smooth inhomogeneous geometry and compare Link-holonomy curvature with the reconstructed continuum curvature; push the load toward saturation and verify the finite radial branch; finally, introduce an inside/outside cut and test the rank-two boundary sector and finite unitary record transfer in the same code base.

Verification stage	Observable	Pass condition
V1. Matrix law	$U^\dagger U - I$ and inverse update	Machine-precision unitary residual
V2. Causality	Support of $U^\dagger O_x U$	No support outside $N1(x)$ per K
V3. Vacuum dispersion	$\omega(k)$	Weyl cone with controlled lattice corrections
V4. Curved principal symbol	Local cone from geometry records	Converges to ADM null cone
V5. Backreaction	Δp_G versus $\partial H_M / \partial G$	Finite identity and smooth-force agreement
V6. Constraint algebra	H-H, H-D, D-D residuals	All vanish on fixed low-mode sector under refinement
V7. Strong gravity	$\rho, \chi(r), m(r)$	Bounded density; regular center; stable branch
V8. Horizon sector	rank, state transfer, reciprocity	M2 support, unitary dimension match, emission/absorption ratio

28. Integrated conclusion

Paper IV v3.0 reframes GCC as one finite microscopic update problem. The fixed graph is not itself curved; geometry is encoded in finite node and frame-Link records. In the uniform sector the same architecture contains the strict-local BCC/Weyl propagator. In a smooth geometry sector, local records control the matter principal symbol and generate an ADM causal cone. The reciprocal action of the same controlled unitary supplies matter

backreaction. A finite-Weyl geometry update with Link-derived curvature has the ADM/DeWitt system as its target infrared universality class, with the DeWitt trace coefficient selected within the stated closure assumptions.

At high load, finite capacity and the reversible Compute-Link circuit prevent unbounded density. The robust prediction is a Planck-density core with cube-root mass scaling; the Hayward profile is the exact minimal uncorrelated branch, while the full correlated circuit gives a bounded deformation. At a semiclassical causal cut, the same bulk carrier yields an orientation-independent rank-two outgoing algebra. Conditional on boundary-support minimality, the resulting binary information sectors have exactly the Hilbert-space dimension needed for unitary transfer to radiation. Microcanonical reversibility then normalizes inclusive emission through the exterior absorption problem.

The architecture is therefore substantially more integrated than the previous revision, but it remains a candidate rather than a completed quantum theory of gravity. The decisive question is now sharply posed: can one explicit finite, autonomous, strict-local many-body matrix rule realize all of these sectors simultaneously and retain the demonstrated infrared behavior under controlled refinement? The verification program of Section 27 is designed to answer that question.

APPENDIX A. Exact finite-Weyl and BCC audit definitions

For a finite Weyl pair, matrix exactness can be monitored by the relative Frobenius residual

$$R_W = \frac{\|W_M W_N - \omega^k W_N W_M\|_F}{\|W_M W_N\|_F}$$

d	Weyl phase exponent	Relative matrix residual
5	2	2.18×10^{-16}
7	4	5.75×10^{-16}
11	8	3.54×10^{-16}
17	14	4.31×10^{-16}

The reported values are floating-point roundoff, illustrating the distinction between exact finite matrix composition and emergent continuum closure.

APPENDIX B. Backreaction identity

Let the finite geometry shift advance the geometry basis by one record and let the matter update be diagonal in the geometry-control basis. Direct multiplication gives

$$C_{GM}^\dagger (X_G \otimes I) C_{GM} = X_G \Omega_G, \Omega_G = \sum_g |g\rangle \langle g| \otimes U_M(g+1)^\dagger U_M(g)$$

No measurement or classical stress extraction is used. In a smooth chart, expand the neighboring controlled unitaries to first order in the geometry-record step and identify the shift phase with the conjugate geometry momentum. The resulting momentum kick is the derivative of the same matter Hamiltonian that controls propagation. Higher-order Baker-Campbell-Hausdorff terms are ultraviolet corrections rather than independent source terms.

APPENDIX C. DeWitt one-half selection

For the general local quadratic kinetic term, only the kinetic-curvature cross bracket contributes second derivatives of the lapses. Contracting the curvature variation with the generalized momentum combination gives a standard momentum-constraint term and a trace-gradient remainder proportional to one minus twice the DeWitt coefficient. Antisymmetrizing the lapse pair and integrating by parts yields the expression in Section 10. If no additional trace constraint is admitted, standard HDA closure therefore selects one half. This is a conditional derivation within the local quadratic ansatz, not a global uniqueness theorem over arbitrary nonlocal or higher-derivative microscopic completions.

$$\{H_\lambda[N], H_\lambda[M]\} = D[\omega] + (1 - 2\lambda) \int d^3x \left(\nabla_a \pi \right) (N \nabla^a M - M \nabla^a N)$$

APPENDIX D. BCC loop curvature and strict locality

The BCC neighbor shell contains independent pairs of edge directions whose cross products span the spatial bivector space. Four-edge loops therefore provide enough independent holonomies to reconstruct the three spatial curvature two-form components in the smooth sector. A direct one-tick plaquette readout would violate radius-one causality because the opposite loop corner is not in the node neighborhood. The reversible scheduler instead transports and multiplies one edge factor per microscopic tick. A completed loop record is cached only after all factors have arrived. The finite delay is derivative suppressed in the infrared and becomes a genuine ultraviolet memory effect near the processing scale.

$$W_{x;s,t} = L_{x,s} L_{x+s,t} L_{x+t,s}^\dagger L_{x,t}^\dagger$$

$$W_{x;s,t} = \exp \left[-\frac{i}{4} \Sigma_{s,t}^{ab} F_{ab}^i \sigma_i + O(\ell_{proc}^3) \right]$$

APPENDIX E. Correlated strong-gravity radial equation

Using the spherical mean-density load, mass conservation and the explicit Compute-Link circuit imply

$$r \frac{d\chi}{dr} = -3\chi(1-\chi) \cos^2(g\chi^2)$$

The uncorrelated limit is logistic and gives the analytic Hayward-type branch. For finite correlation the equation is separable in principle but not generally elementary. The endpoints remain fixed because the circuit correction vanishes at zero and unit load. The weak-load expansion preserves the quadratic leading term, while the correlation correction starts at fifth order. Hence the central density cap and cube-root mass scaling are more robust than the detailed exterior coefficient.

APPENDIX F. Physical-state construction at microscopic level

The microscopic gauge problem is separated from the emergent Hamiltonian constraint problem. Internal frame rotations and address relabelings act as finite redundancies of the record description. Projecting onto their invariant sector gives a candidate microscopic physical Hilbert space.

$$P_{phys} = P_{addr} P_{frame}, H_{phys}^{micro} = P_{phys} H, [U_{GCC}^{full}, P_{phys}] = 0$$

A nonzero invariant vacuum guarantees that the projected space is not empty. Nontrivial gauge-invariant excitations can be generated by closed Link holonomies and relational matter combinations. What remains open is whether the full nonlinear 3+1 QCA admits a sufficiently rich invariant sector whose refined relational observables reproduce the continuum physical state space of gravity plus matter.

APPENDIX G. Horizon dimension matching and information transfer

The binary boundary algebra implies exponential horizon information growth with the number of independent records. Removing a finite number of records while storing the same records in radiation preserves the total Hilbert-space dimension exactly.

$$\dim H_H(n) = 2^n = 2^{n-q} 2^q = \dim [H_H(n-q) \otimes H_R(q)]$$

This equality is the algebraic reason a lossless finite-dimensional transfer is possible. The explicit map in Section 21 is a convenient bijection; the full interacting QCA may realize a more complicated energy- and channel-dependent unitary within the same dimension-matched block. Microcanonical reciprocity depends only on reversibility and state counts, so the inclusive normalization survives that microscopic freedom.

APPENDIX H. Logical hierarchy of the v3.0 construction

Level	Exact requirement	Emergent / conditional requirement
Microscopic algebra	Finite matrices; exact unitarity; finite Weyl composition	Canonical variables only in low-phase charts
Microscopic causality	One-neighborhood support per K	Continuum light cone in low-wave-number sector

Level	Exact requirement	Emergent / conditional requirement
Gauge	Finite frame/address covariance	Continuum diffeomorphism interpretation under refinement
Matter	Local QCA propagation	Curved Weyl/Dirac QFT and interacting continuum fields
Gravity	Finite geometry/Link update	ADM/DeWitt HDA and Einstein-Hilbert action
Strong gravity	Finite capacity	Regular spherical branch and detailed profile
Horizon information	Rank-two outgoing carrier algebra	Boundary-support minimality and area matching
Evaporation	Finite unitary block transfer	Greybody spectrum, line broadening, full interacting Page dynamics

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